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University of Iowa

Library Policy: Use of Generative Artificial Intelligence in Metadata Creation

2nd October 2024

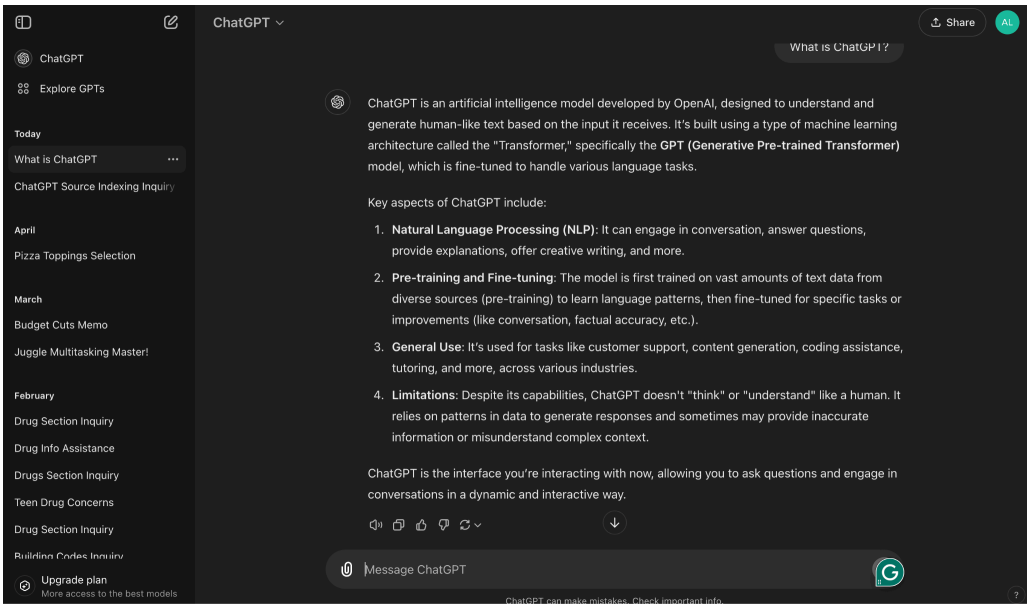
INTRODUCTION

AI is a useful tool that improves efficiency. But it should not be forgotten that human experiences and discernment capabilities have value, especially in information professions that focus on serving human patrons. This policy document guides all technical services staff members to ensure that generative AI is used appropriately to enhance the quality and accessibility of metadata while maintaining the trust and integrity of the library's collections.

APPROVED AI PLATFORMS

ChatGPT and ChatGPT+ by OpenAI

Chat GPT (free tier) and Chat GPT+ (subscription tier) can mine large documents without indexing source files. ChatGPT+ provides access to GPT-4, an advanced model with an improved knowledge base and stronger ethical guidelines. Both models are trained with human feedback and continuously learn from incoming queries.



ChatGPT User Interface.

ALLOWED USES

AI may be used to create bibliographic metadata in standards such as MARC and Dublin Core. AI is best suited for technical metadata. Descriptive metadata should be manually created or carefully verified to combat hallucinations and errors.

- Generative AI may be used for technical fields, including:
    - Format
    - File size
    - Identifier
    - Source
- Generative AI may be used for the following fields only if verified (see Staff Workflow verification procedure):
    - Contributor
    - Coverage
    - Creator
    - Date
    - Description
    - Language
    - Publisher
    - Relation
    - Rights
    - Source
    - Subject
    - Title
    - Type

STAFF WORKFLOW

How to Use AI for Metadata Creation

Input

To create a complete metadata record, input explicit prompts in the generative AI interface. Prompt examples:

- Create a complete Dublin Core metadata record formatted as a table for [Title] by [Author Name], published [Publication Date].
- Create a complete Dublin Core metadata record in XML format for [Title] by [Author Name], published [Publication Date].

- Create a complete MARC record using RDA for [Title] by [Author Name], published [Publication Date].

Note: The specific fields included in prompts may vary.

Verification Procedure

Generative AI is susceptible to “hallucinations”—mistakes or made-up information. The library’s verification procedure ensures records utilizing AI-generated metadata outputs are accurate, comprehensive, and unbiased. Complete the following steps for the verification procedure:

- Step 1: Manually review AI-generated metadata records.
- Step 2: Edit any inaccuracies.
    - Pay special attention to the **Description** field. AI-generated descriptions should be compared to the publisher-provided description.
    - Note inaccuracies may be more frequent with materials published within the last year.
- Step 3: Edit any placeholder text in fields that are collection-specific.

How to Cite AI Created Metadata

When partially AI-generated documents enter our collection, staff must acknowledge the use of AI in their creation. Look in the acknowledgments section to determine whether a document contains AI-generated content. For scholarly works, also check in the methods section.

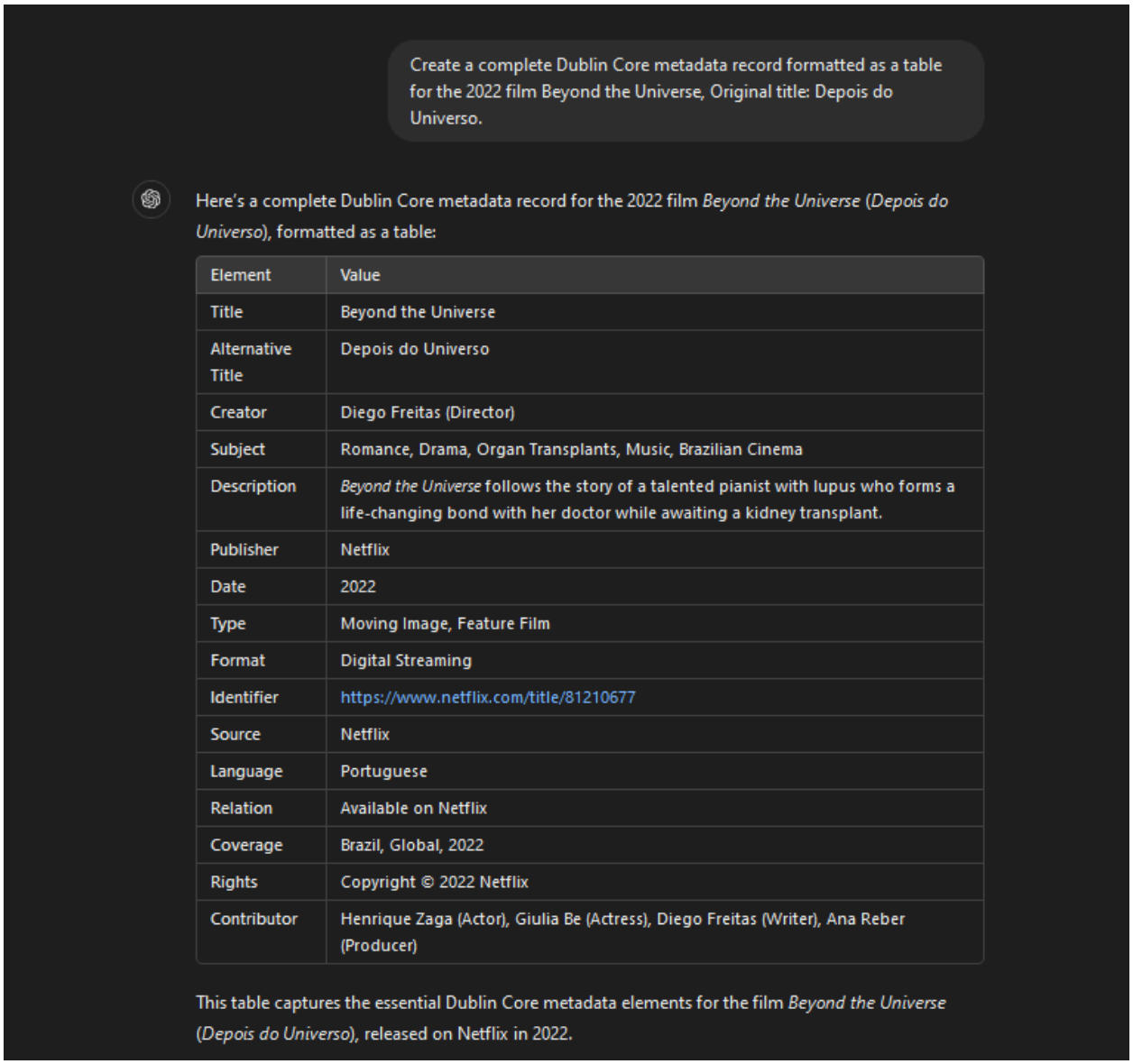
Indicate the use of a generative AI model in the **Contributor** field. Include the following in your entry if known:

- The name of the generative AI model.
- The version information.
- The organization that created the generative AI model.

For example:

Contributor	ChatGPT-4o, OpenAI

Example Cases



Create a complete Dublin Core metadata record formatted as a table for the 2022 film Beyond the Universe, Original title: Depois do Universo.

Here's a complete Dublin Core metadata record for the 2022 film *Beyond the Universe* (*Depois do Universo*), formatted as a table:

Element	Value
Title	Beyond the Universe
Alternative Title	Depois do Universo
Creator	Diego Freitas (Director)
Subject	Romance, Drama, Organ Transplants, Music, Brazilian Cinema
Description	*Beyond the Universe* follows the story of a talented pianist with lupus who forms a life-changing bond with her doctor while awaiting a kidney transplant.
Publisher	Netflix
Date	2022
Type	Moving Image, Feature Film
Format	Digital Streaming
Identifier	https://www.netflix.com/title/81210677
Source	Netflix
Language	Portuguese
Relation	Available on Netflix
Coverage	Brazil, Global, 2022
Rights	Copyright © 2022 Netflix
Contributor	Henrique Zaga (Actor), Giulia Be (Actress), Diego Freitas (Writer), Ana Reber (Producer)

This table captures the essential Dublin Core metadata elements for the film *Beyond the Universe* (*Depois do Universo*), released on Netflix in 2022.

Example Dublin Core metadata record created with ChatGPT-4o.



Example Dublin Core metadata record in XML format created with ChatGPT-4o.

Create a complete MARC record using RDA for This Is How You Lose the Time War by Amal El-Mohtar and Max Gladstone published in 2019.

Here's a complete MARC record using RDA (Resource Description and Access) standards for *This Is How You Lose the Time War* by Amal El-Mohtar and Max Gladstone, published in 2019:

swift Copy code

```
=LDR  00000nam a2200000 i 4500
=001  tihtltw_2019
=003  DLC
=005  202309xx
=008  190701s2019    nyu           000 1 eng d
=020  ##$a9781534431003$q(hardcover)
=020  ##$a9781534431010$q(ebook)
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=250  ##$aFirst edition.
=264  #1$aNew York, NY :$bSaga Press,$c2019.
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=338  ##$avolume$bnc$2rdacarrier
=490  1#$aThe Time War series ;$vbook 1
=500  ##$aA novel blending elements of science fiction, romance, and epistolary narrati
=520  ##$aIn the midst of a war between rival time travelers, two agents begin to excha
=650  #0$aScience fiction.
=650  #0$aTime travel$vFiction.
=650  #0$aEpistolary fiction.
=650  #0$aSpace and time$vFiction.
=650  #0$aRomance fiction.
=700  1#$aGladstone, Max,$eauthor.
=830  #0$aThe Time War series ;$vbook 1.
=856  41$uhttps://www.simonandschuster.com/books/This-Is-How-You-Lose-the-Time-War/Amal
```

Example MARC metadata record created with ChatGPT-4o.

BIBLIOMETRIC ANALYSIS & EFFECTIVENESS

Generative AI may also be used for bibliometric analysis. Its effectiveness for this purpose is neatly summarized in librarian Deep Kumar Kirtania's paper, *ChatGPT as a Tool for Bibliometric Analysis: Interview with ChatGPT.* Kirtania asked ChatGPT the following three questions:

1. "How ChatGPT can help to conduct a bibliometrics analysis?"
2. "How ChatGPT can help a bibliometrician?"
3. "What are the challenges associated with ChatGPT while conducting a bibliometrics analysis?"

ChatGPT responded to Kirtania's first question by noting that it can collect, process, clean, and analyze data, as well as provide recommendations and interpretations of bibliometric data.

Responding to Kirtania's second question, ChatGPT stated it can retrieve raw data from various sources, preprocess raw data, analyze data, create visualizations of data, generate literature reviews, and build recommender systems to suggest additional material.

Answering Kirtania's final question, ChatGPT admitted five challenges including its limited domain knowledge, potential inaccuracies in its data (especially when handling large datasets), the influence of various biases, need for human interpretation, and ethics/privacy considerations.

Kirtania's findings are included in this policy guide to remind library staff that while ChatGPT has remarkable capabilities, it also has several limitations. This emphasizes the importance of following our library's verification procedure and keeping the next section's ethical and legal implications in mind.

ETHICAL AND LEGAL IMPLICATIONS

Privacy and Data Protection

AI-powered language models, like ChatGPT, collect and retain user inputs to improve performance. These models therefore come with privacy risks. Privacy is a fundamental value in librarianship, as noted in both the Library Bill of Rights and ALA Code of Ethics. In an effort to protect each library user's right to privacy and confidentiality, staff may not enter the following information into an AI tool:

- Personal patron information (i.e., name, address, email, materials borrowed)
- Intellectual property (i.e., your own scholarship or research, published content)
- Confidential information

Staff should be transparent with patrons about the library's generative AI use, openly communicate the library's AI initiatives, and invite patron feedback where applicable.

Legal and Compliance

All AI use must comply with relevant state and federal law.

AI-generated outputs is an unsettled area of law, and AI tools sometimes use copyrighted material without a license. Staff must be careful in their reproduction of outputs and approach all AI uses with ethical consideration for its impact on library services, resources, patrons, and staff.

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